

Cardiac Risk Prediction

An explainable AI framework for biomarker-driven cardiovascular risk stratification.

Model: CatBoost (Optuna-tuned) — 97% accuracy, AUC 0.9973

Patient Info

Patient Name

usha

Age

45

-

+

Upload Lab Report (CSV)

Upload a CSV with patient lab values (one row)

Drag and drop file here
Limit 200MB per file • CSV

Browse files



sample_patient.csv 0.6KB



File loaded. Values auto-filled below (editable).

Key Lab Values

Standardized z-scores: 0 = population average, positive = above average, negative = below average.

c_reactive_protein

-0.5771

-

+

troponin_level

-0.0644

-

+

fasting_glucose

-0.0464

-

+

creatinine

1.0159

-

+

cholesterol

troponin

crp_hdl_ratio

troponin_crp_product

0.0167

-

+

-0.4116

-

+

platelet_count

wbc_count

0.1820

-

+

0.8438

-

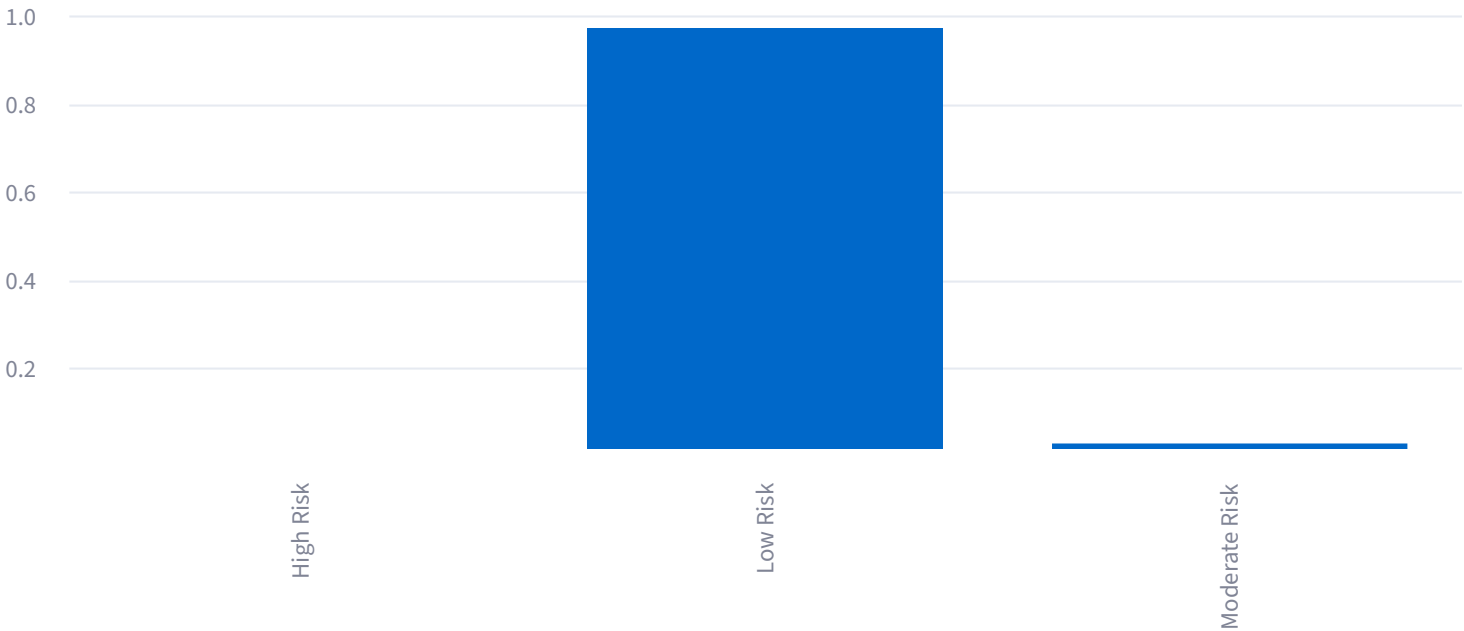
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Predict Risk

Result for usha (Age 45)

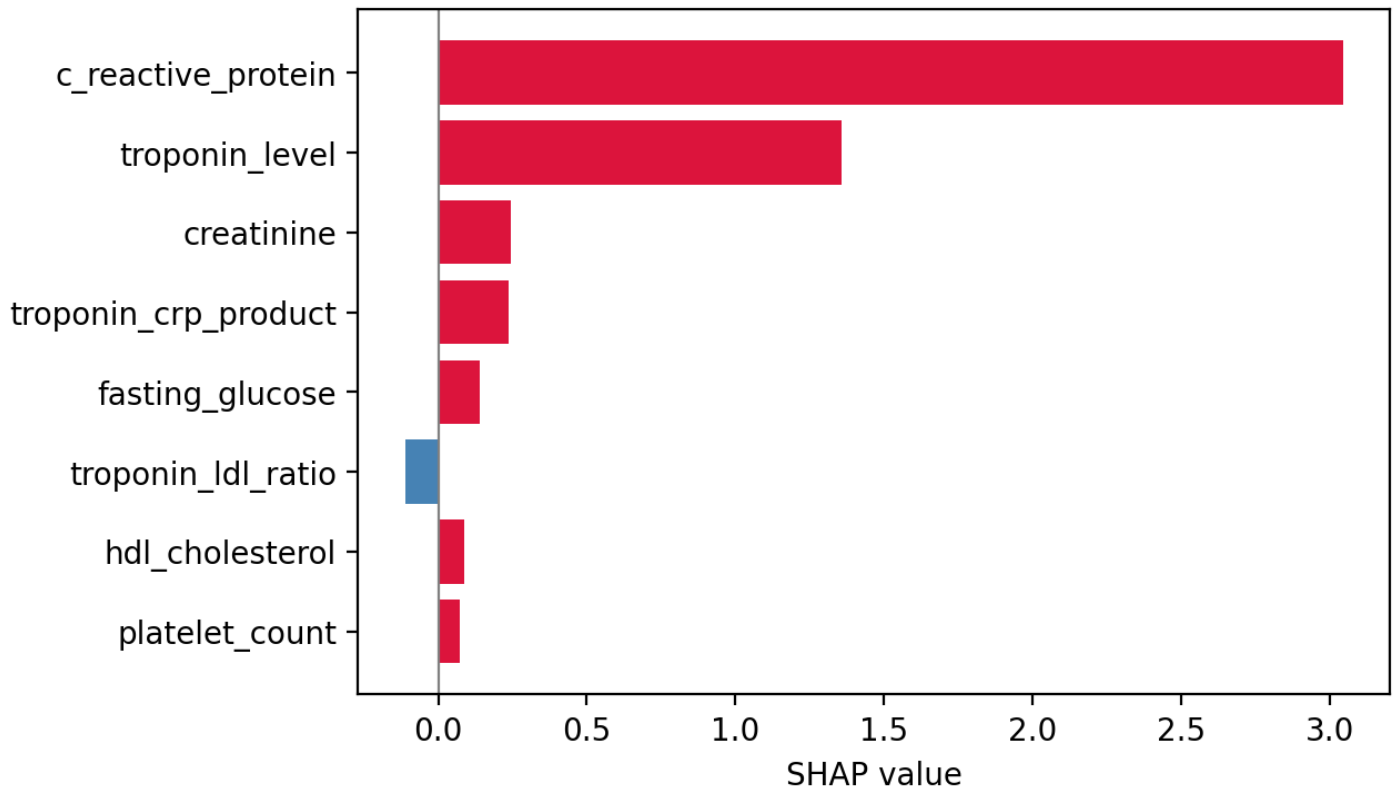
Predicted Category: Low Risk

Prediction Confidence: 97.06%



Key Factors Behind This Prediction

Key Factors Behind This Prediction



Red = pushes toward predicted risk. Blue = pushes away from it.

Patient Risk Assessment Report

Predicted Risk Category: Low Risk

Prediction Confidence: 97.06%

Primary Contributing Biomarkers:

1. **c_reactive_protein** (positively contributed; SHAP impact = 3.0452)
2. **troponin_level** (positively contributed; SHAP impact = 1.3556)
3. **creatinine** (positively contributed; SHAP impact = 0.2425)
4. **troponin_crp_product** (positively contributed; SHAP impact = 0.2363)

Model Interpretation:

The predicted risk classification is driven by the cumulative effect of the above biomarkers. Features with larger absolute SHAP values exert stronger influence on the final model decision. Comparatively weaker feature contributions reduced the likelihood of alternative risk categories.

 **Clinical Note:** This output represents a probabilistic estimate generated by a machine learning model. It is intended for decision-support purposes only and must not replace comprehensive clinical evaluation. All medical decisions should be made by a qualified healthcare professional.